ASSET BIBLE · BOMB · VFX · UI · WORLD · AUDIO

FUSE VALLEY

Premium or nothing. Every asset, specified to the Quality Bar.



DOCUMENT
Asset Bible

VERSION
1.0 · 2026-10-17

PLATFORM
Roblox · PC · Mobile · Console

BUILT WITH
Rojo · Luau (strict) · Claude Code

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> Every model, texture, effect, sound and screen in FUSE VALLEY is specified here. The user's standard is the rule this document enforces: **"EVERY ASSET, EVEN THE BOMB, HAS TO BE EXTREMELY RICH, PREMIUM AND HIGH QUALITY. ABSOLUTELY ZERO AI SLOP."**

0.1 Document map

§	Section
1	The Quality Bar (read before making anything)
2	Art direction: "Arcade Pyrotechnic"
3	The Bomb (hero asset) and its skins
4	World: maps, lobby, props, lighting
5	VFX: explosions, fuses, defuses, feedback
6	UI: system, HUD components, callouts, icons, motion
7	Screens & sequences
8	Launch cosmetic catalogue
9	Audio
10	Pipeline: authoring, review, import, naming

1. The Quality Bar

1.1 What "premium" means here

An asset is premium when **someone who has never seen the game could screenshot it, and the screenshot would sell the game.** In practice, every asset has all of these:

1. **Intent:** it was designed for this game, for a specific moment, by a decision someone can explain.
2. **A strong silhouette:** it reads as itself in pure black at 40 px tall.
3. **Material truth:** iron looks like iron, brass like brass, ceramic like ceramic. Surfaces have wear, edges catch light, and nothing is flat plastic by default.
4. **Three reads:** it works far away (shape and colour), mid-range (main details) and close up (fine detail, wear, texture).
5. **Life:** it moves, reacts or responds. The bomb breathes, buttons squash, numbers count, lights pulse.
6. **Sound pairing:** anything that moves or changes state has a matching sound designed for it.
7. **Consistency:** it uses the palette, materials, lighting and motion rules of §2. One asset done "differently" breaks the whole look.

1.2 Banned ("AI slop" and its cousins)

Never	Why	Instead
AI-generated images, textures, icons, 3D or audio, as finals *or* placeholders	Generic, inconsistent, and players spot it instantly	Hand-authored procedural Blender scripts, hand-drawn vectors, the licensed Roblox audio library, layered and processed
Toolbox or free models	Wrong style, unknown scripts, other people's work	Our own models (§10)
Default `SmoothPlastic` everywhere on hero assets	Reads as unfinished	PBR `SurfaceAppearance` (colour, normal, roughness, metalness maps)
Emoji as icons	Inconsistent across platforms; looks cheap	The custom icon set (§6.4)
Generic two-stop gradients as decoration, glassmorphism, frosted panels	The "template UI" look	The panel system in §6.2: solid layered fills, bevel highlight, crisp ink outline
Default fonts (Arial, SourceSans, Gotham as display)	Looks like a default game	The type scale in §2.6
"Coming soon" tiles, lorem ipsum, placeholder text, dead buttons	Broken promise	Ship fewer items, all finished
Static pop-in UI	Feels dead	Every appear, press and change is animated (§6.5)
Unmatched stock sounds (one generic "click" for everything)	No identity	A designed sound for each action (§9)
Stretched, blurry or upscaled textures	Looks cheap on retina phones	Authored at 2× display size, power-of-two
Particle soup (huge counts, no shape)	Noisy and slow	Designed stacks with timing (§5); budgets in GDD §24.2

1.3 The checklist — every asset passes before it ships

#	Check	How
1	Designed for a named moment in the GDD	Cite the GDD feature ID in the asset manifest
2	Silhouette test	A black fill render at 40 px tall is recognisable
3	Three-distance test	In-game captures at 6, 25 and 60 studs
4	Phone test	A 844×390 Studio emulator capture; detail and text legible
5	Palette	Only §2.3 tokens or their documented tints
6	Materials	PBR maps present; roughness varies across the surface (no uniform values)
7	Wear and imperfection	Edge wear, soot, scuffs or hand-made irregularity where the material would have it
8	Lighting	Checked under all 3 lighting presets (§4.4)
9	Motion	Idle or response animation defined and implemented
10	Sound	A paired sound exists and is mixed (§9)

#	Check	How
11	Budgets	Tris, parts, textures and particles within GDD §24.2
12	Readability rules	GDD §10.4 holds (team colours, fuse truth, silhouette)
13	Review renders	A turntable contact sheet in `art/out/<Asset>/`
14	Hero sign-off	The bomb, the explosion, the logo, the lobby and the launch map need the **user's explicit approval**

An asset that fails any check does not ship, not even as a placeholder. A placeholder is a *simpler* version of the final asset that passes this checklist, **never** a lower-quality one.

2. Art direction: the reference-video look ("Toy Arcade"), 2026-10-17

> **Supersedes the original "Arcade Pyrotechnic" direction where they conflict** (DECISIONS 2026-10-17: the owner's reference video wins for UI layout, colours, buttons, motion and world brightness; the fireworks / bomb theme stays). The rest of this section (2.1–2.7) is the original direction, kept for the bomb's materials and everything the video doesn't cover.

Area	The rule now	Where it lives
World	Bright, sunny, toy-like: studded plastic surfaces (Theme.StudTexture), saturated toy colours, palms and voxel trees, puffy clouds, crisp midday light	`builders/MapYardToy.luau` (the Yard restyle pass), concept `art/ref/map/yard_concept_overview.png`
UI panels	Flat rounded bodies with a soft vertical gradient and a gloss band, a coloured border ring, a thick ink outline; screen headers are a gradient strip (rainbow for the Shop) with the title in big round type and its 3D icon breaking out over the corner; a red square close button	`src/client/UI/Skin.luau`, `DailyController.panel`
Buttons	Bright arcade buttons: lighter top, darker lip, gloss band, ink outline, white text with an ink stroke; green = buy / play, red = close, a colour per tab	`Skin.button`, `Skin.Preset`
Cards	Rarity-gradient cards (Common grey-white, Rare blue, Epic purple, Legendary gold, Exclusive pink) with a white border, halftone wash, the item's art large	`StoreController`
Type	Fredoka One everywhere (display, numbers, labels), Builder Sans Bold only for long body copy; Luckiest Guy for the logo	`Theme.Font`, `Skin.Font`
Icons	Glossy, chunky 3D icons modelled and rendered in Blender (art/icons/set_*.py) after ChatGPT style sheets (art/ref/icons — references only)	`Assets.Icon`
HUD	Left dock of square tiles (icon, name, shortcut key, "!" badge), the Stars wallet bottom-left (under the top bar on phones), the gear menu in the top-bar row	`UI/Dock.luau`, `ProgressController`, `SettingsController`
Screens	Video-style join screen (logo + PLAY!), STARTER PACK rainbow card, crate opening with rays and confetti	`src/first`, `OfferController`, `CrateController`
Sound	The in-house UI pack and SFX set (art/audio/make_ui.py, make_sfx.py) + the licensed library	`Assets.SFX`

The Quality Bar (§1) is unchanged: no AI-generated finals, no Toolbox models, everything crafted, animated and paired with a sound. "Generic two-stop gradients" stay banned *as decoration on dark panels*; the reference's bright gradient cards, tabs and headers are the house style now.

2b. The original direction: "Arcade Pyrotechnic"

2.1 One sentence

A hand-crafted fireworks workshop at golden hour: warm iron, polished brass, glazed ceramic and braided fuse cord, lit by sparks. It's stylised and chunky like a premium arcade game, but every surface is made of a real material.

2.2 Pillars

Pillar	Means	Doesn't mean
Chunky & readable	Big, simple primary shapes. 60 % big forms, 30 % medium, 10 % small detail. Rounded bevels everywhere (no razor edges).	Low effort or low detail
Material-rich	Iron, brass, copper, ceramic, painted wood, canvas, rope, glass: each with PBR maps and wear	Realism. Proportions stay stylised
Warm vs cool	Warm light from fire and sparks against cool dusk shadows. The bomb is always the warmest, brightest thing near it	Muddy all-orange scenes
Crafted, not manufactured	Rivets, stitching, stencils, hand-painted signage, chalk marks, tool marks	Sterile "clean 3D"
Pyro joy	Explosions are celebrations: colourful, shaped, satisfying	Violence, gore, grim realism

2.3 Palette (tokens) — `Theme.Palette`

Token	Hex	Use
`Ink`	`#16121C`	Outlines, darkest UI, text on light
`Ink2`	`#241D2E`	Panels (dark)
`Smoke`	`#3B3447`	Panel bevels, disabled
`Ash`	`#8C8498`	Secondary text
`Paper`	`#FFF3E2`	Light panels, body text on dark
`Fuse`	`#FF6A1F`	**Brand primary**: logo, primary buttons, fuse
`Ember`	`#FF9442`	Highlights on Fuse
`Spark`	`#FFD84A`	Sparks, rewards, Stars
`Gold`	`#F2B13A`	Premium, VIP, Legendary
`Detonate`	`#E8322B`	Danger, the CRITICAL stage, boom
`Runner`	`#2FD3FF`	**Reserved:** the Runner team
`Bomber`	`#FF4A2E`	**Reserved:** the Bomber team
`Safe`	`#3BE48C`	Safe Strips, success, defuse
`Mega`	`#B04CFF`	The MEGA BOMB event
`Iron`	`#2E2C33`	Bomb body base

Token	Hex	Use
`Brass`	`#C9973F`	Bands, trims
`Porcelain`	`#F4EFE6`	Fuse socket, ceramic

Rarity frames: Common `#A9B1C2` · Rare `#33B6FF` · Epic `#A65BFF` · Legendary `#FFB627` · Exclusive: obsidian `#1E1A24` with an iridescent foil texture.

2.4 Materials library (`art/lib/materials.py` → SurfaceAppearance sets)

Material	Colour	Roughness	Metal	Signature detail
Cast iron	`Iron` with 6 % hue noise	0.55–0.8	0.85	Hammered dimples (normal), soot gradient toward the top, polished wear on the equator
Brass	`Brass`	0.25–0.45	1.0	Edge brightening, fingerprint smudges (roughness), darker patina in recesses
Copper	`#B8643A`	0.3–0.5	1.0	A verdigris tint in crevices
Glazed porcelain	`Porcelain`	0.08–0.15	0	Crackle-glaze normal, a slight blue cast in the glaze pools
Braided cord	`#D9B98A` → charred `#2A1F1A`	0.85	0	A real braid pattern in normal and colour; charring along the burn
Painted steel	Any token	0.4–0.6	0.3	Chipped paint at edges showing the metal under it
Canvas & leather	`#6B4A32`	0.7–0.9	0	Stitching, stretch creases
Glass lens	Tinted	0.05	0	An inner emissive element (a Neon sub-part)

Texture size: **1024²** for hero assets (the bomb, the lobby centrepiece), **512²** for props, and trim sheets for architecture.

2.5 Lighting presets (`Config.Lighting`)

Preset	Use	Setup
Golden Fuse (default)	Maps and lobby	Future lighting; the sun is warm `#FFC58A` at a low angle (ClockTime 17.4); cool ambient `#5D6A8F`; Atmosphere density 0.32 with a warm haze; Bloom 0.6 / threshold 1.6; ColorCorrection saturation +0.08, contrast +0.06
Night Show	MEGA BOMB rounds	ClockTime 19.8, deep blue ambient, every emissive and spark brightened, fireworks in the skybox
Spotlight	Last Stand	Global brightness × 0.35, a SpotLight on the Runner, a warm rim light on the Bombers

2.6 Typography (Roblox fonts)

Role	Font	Weight	Use
Display / callouts	**Luckiest Guy**	Regular	CLUTCH DEFUSE!, VICTORY, titles, logo lockups
UI headings & numbers	**Builder Sans**	ExtraBold / Heavy	Counters, prices, tabs, timers (tabular figures)

Role	Font	Weight	Use
Body	**Builder Sans**	Medium / Bold	Descriptions, tips

Rules:

- Display text always has a 3 px `Ink` stroke plus a 4 px drop shadow (`Ink` at 40 %), offset straight down. Never a glow-only outline.
- Mobile minimum sizes: body 14 px, numbers 16 px, buttons 18 px.

2.7 Shape language

- **Circles = the bomb, danger, timers.** Rings, fuses, dials.
- **Rounded rectangles with a chunky bevel = UI and safe things.**
- **Diagonal hazard stripes = the field**, warnings and the Field Fuse.
- **Starbursts = rewards**, only on reward moments, so they stay special.

3. The Bomb — "The Tick" (hero asset)

The bomb is on screen more than anything else in the game, and it's the face of the brand: logo, icon, thumbnails. **It gets more care than any other asset.**

3.1 Design brief

> A hand-forged cast-iron fireworks bomb from an old pyrotechnics workshop: heavy, round and honest. A riveted brass belt with two glass indicator lamps, a glazed porcelain fuse socket, and a thick braided fuse that really burns. When it's about to blow, the welded seams glow white-hot from inside, as if it can barely hold the blast in.

Why this design:

- The round black bomb is the most universally readable "bomb" shape there is. We keep that instant read (pillar P1) and make it premium through **materials and mechanisms**, not a weird shape.
- The lamps, glowing seams and burning fuse give us **three independent timer signals** that work at any distance:
  - **Far:** the colour of the seam glow
  - **Mid:** the lamp pulse rate
  - **Close:** the fuse length

3.2 Anatomy & dimensions (for a 5-stud R6 character)

Part	Spec	Size (studs)	Material
Body	A slightly squashed sphere (Y × 0.96), 24 rings, smooth-shaded. Subtle hand-forged irregularity (±1.5 % radial noise, low frequency).	Ø 2.2	Cast iron (§2.4): hammered dimples, a soot gradient on the top third, polished wear on the belt line
Belt	An equatorial band, slightly proud of the surface, with chamfered edges	0.30 tall, 0.05 proud	Brass: edge brightening, darker patina between the rivets
Rivets	14 domed rivets, evenly spaced on the belt	Ø 0.09	Brass, brighter polished caps
Indicator lamps	2 glass lenses (front and back) in brass bezels. Each has an inner Neon element whose colour and brightness pulse with the tick	Ø 0.26	Glass lens + Neon core (`Spark` → `Detonate` by stage)
Seams	2 meridian weld seams (an X when seen from the top), sunk 0.02 into the body. A separate **SeamGlow** MeshPart (an inset strip) carries the glow	0.05 wide	Body: iron with a weld bead normal. SeamGlow: Neon, transparency driven by stage
Collar	A brass neck ring with a knurled band	Ø 0.62, 0.22 tall	Brass, knurl normal
Fuse socket	A glazed porcelain cup sitting in the collar	Ø 0.42, 0.18 tall	Porcelain with crackle glaze; a soot ring at the rim
Fuse	A braided cord in **12 segments**, gently curved (an S-bend), tapering 0.11 → 0.08	1.25 long at full	A braided cord texture (real 3-strand braid normal), charring along the burn

Part	Spec	Size (studs)	Material
Spark	At the burning tip: a ParticleEmitter, a PointLight and a tiny bright core sprite	—	§5.2
Stencil	A small stencilled "FV" mark and the hazard pictogram on the back, low contrast (painted, chipped)	0.35	A decal baked into the colour map
Clamp	A sprung steel claw that grips the bomb, with a **visible coil spring** to a small leather-and-buckle back plate	Plate 0.9 × 1.1	Painted steel (`Ink2`, chipped), leather, a brass buckle

Mount: the back plate sits on the Torso back, 0.4 studs out. The bomb hangs on the claw, centred 0.15 studs above the torso's centre line.

3.3 States & behaviour (implemented in `BombController`)

State	Duration	Visual	Audio (§9)
Stick	0.22 s	The bomb slams in from the tagger's hand direction. Squash 1.15 × 0.85 → overshoot → settle. The claw snaps shut, and a dust puff comes off the back.	`Bomb.Clamp` (a metal clank) + `Bomb.Ignite` (fuse catches)
LIT (4.0–2.5 s)	—	The spark is steady and the lamps glow `Spark` with a slow sine pulse (2 Hz). The seams are dark. **Secondary motion:** spring jiggle driven by the character's acceleration (k = 180, damping 0.35), ±8° wobble.	`Bomb.Tick` at 2 Hz, `Fuse.Hiss` loop
HOT (2.5–1.0 s)	—	The seams fade in orange (`Ember`, transparency 0.7 → 0.3). The lamps turn orange and pulse at 4 Hz. Smoke wisps rise from the fuse socket.	Tick at 4 Hz with pitch +2 semitones; the hiss gets louder
CRITICAL (1.0–0 s)	—	The seams go white-hot (`Spark` → white core). The lamps turn `Detonate` and strobe at 10 Hz. **The body swells 8 % and shakes** (±0.04 studs). Sparks spray from the seams, a PointLight (range 10, orange) switches on, and heat shimmer appears if the device allows.	Tick at 10 Hz rising into a whine (`Bomb.Critical`), plus the heartbeat (Carrier only)
Pass flight	0.18 s	The bomb pops off the back, flies a 2-stud-high arc with 1.5 spins and a short warm motion trail, then lands on the receiver (Stick, pitched up).	`Bomb.Whoosh` + `Bomb.Clamp` (pitched +3 semitones)
Boom	—	A pre-swell over 0.08 s (× 1.25), then the explosion stack (§5.1). The bomb mesh is hidden at the flash; brass fragments are part of the debris.	`Boom.*` stack
Defuse	0.9 s	The fuse snuffs (the spark dies with a steam curl). The lamps fade to dark, the claw releases with a click, and the bomb drops, bounces twice with a heavy iron clunk and rolls, then fades out after 1.5 s.	`Fuse.Snuff` + `Bomb.Release` + `Bomb.Drop`
Locked (on a stunned Bomber)	—	Same stages, but the claw is red-painted to show it can't be passed	—

3.4 The fuse: how it burns (technical)

- The 12 segments are separate small MeshParts, welded in a chain.
- **Burn progress** `p = 1 - remaining / FUSE`:
  - Every segment with an index below `floor(p × 12)` is hidden.
  - The current segment scales along its length by the fractional part, and its colour lerps to charred.

    - The spark attachment follows the current tip.

- **This makes the fuse length always the true time** (GDD §10.4).
- **A fresh bomb from a pass keeps its burn state.** The fuse is short on arrival, which visibly tells the receiver "you have no time".

3.5 The overhead fuse ring (UI, every Carrier)

- A BillboardGui ring above the head: 64 px on phones, 72 px on PC. AlwaysOnTop within 60 studs.
- **Its design:** a circular braided-fuse ring that burns away clockwise, with a spark at the burn point. The remaining arc is coloured by stage (`Spark` → `Ember` → `Detonate`).
- **At CRITICAL** the ring shakes and pulses.
- **The skin's Fuse Spark cosmetic** changes the spark sprite, never the arc colours.

3.6 Skins — rules

1. **Same mesh, same silhouette, same parts.** Skins are *texture sets* (colour, normal, roughness, metalness) plus a lamp colour and an optional emissive accent. One SurfaceAppearance swap per part; the geometry is shared.
2. **The fuse, seams and lamp stages are never changed** by a skin. The stage colours always read the same.
3. **No dominant Runner cyan or Bomber red-orange** on the body.
4. **Every skin passes the Quality Bar on its own**, with turntable renders.
5. **The face rig (§3.8) is shared by every skin.** Skins never paint eyes or mouths on the front face zone.

3.7 Launch skin line-up (16)

Skin	Rarity	Concept & material notes
Classic Iron	Default	The hero design above
Rookie Brass	Rare (Starter Bundle)	A full brass body with copper rivets, polished; gentle fingerprints
Copper Kettle	Common	Hammered copper with verdigris in the rivets; a tea-kettle charm
Delft	Rare	Glazed white porcelain with hand-painted cobalt-blue floral patterns (vector-painted, then baked); gold belt
Watermelon	Common	A glossy rind-green body with stripes, a pink "flesh" belt with seed rivets
Gumball Machine	Rare	A clear glass body (two-layer: a glass shell plus a coloured gumball cluster inside), a red-painted metal belt
Lava Forge	Epic	Obsidian body with magma cracks (an emissive accent along the crack geometry baked in the normal), pulsing slowly
Frostbite	Epic	A frosted-ice body with an inner rime layer, a silver belt, breath-frost particles
Clockwork	Epic	A brass and glass body showing turning gears inside (an animated inner part), a dial on the belt
Thunder Egg	Epic	A geode shell cracked open on one side showing amethyst crystals (normal and colour), stone texture
Disco Night	Rare	A mirror-tile body (a mosaic normal plus high metalness) that catches the world light
Paper Lantern	Common	Red lacquered paper over ribs, a gold-leaf belt, a soft inner glow

Skin	Rarity	Concept & material notes
Galaxy	Legendary	Deep space inside a glass sphere: parallax nebula layers (two nested shells), drifting star particles, a chrome belt
Dragon Egg	Legendary	Scaled, iridescent shell plates (normal plus a colour shift), a gold claw belt; the lamps are dragon eyes that blink
Gilded	Exclusive (VIP)	Gold leaf over iron with engraved scrollwork; champagne-gold lamps
Streak Bomb (monthly)	Exclusive (7-day streak)	A new design every month (October: **Jack-o'-Fuse**, a carved pumpkin with candle glow inside)

3.8 The face rig (character) — awaiting the user's sign-off

Brief: the bomb is the star of the game, so it gets a personality. It is cocky while it ticks, nervous at HOT, and screaming at CRITICAL, and it watches its next victim. Players should feel that the bomb *wants* to go off on someone, and the Carrier should see it panicking on their own back.

Placement. The face sits on the front (+Y, away from the clamp), between the two front weld seams. The eyes are above the brass belt and the mouth below it, so the front indicator lamp sits between them like a little glowing nose. Because the bomb hangs facing backward, everyone behind the Carrier sees the face, including the Carrier's own third-person camera.

Part	Spec	Material	Moves
Eyes	2 domes, Ø 0.52, centred 21.5° either side of the front and 26° above the belt, 0.09 proud	Porcelain (the fuse socket's crackle glaze)	Fixed
Pupils + glints	An ink cap (25° of the eye) with a 6.5° porcelain glint	Ink enamel, emissive glint	Rotate about the eye centre (gaze)
Lids	An upper half-shell over each eye, 0.016 thick	Ink enamel	Rotate about the eye's horizontal axis (blink, squint) and its out axis (slant)
Brows	Forged bars, 0.46 long and 0.115 tall, tapering, at 47.5° elevation	Brass (the belt's)	Rotate about the body centre (raise, tilt)
Mouths	One mesh per shape, wrapped onto the body 33° below the belt: **Smirk** (LIT), **Grimace** (HOT, clenched porcelain teeth), **Panic** (CRITICAL, a white-hot core matching the seams), **Ooh** (pass flight), **Grin** (a tag)	Brass lips, porcelain teeth, emissive core	Shown one at a time

Expressions (driven by `BombController` from the bomb's state):

Moment	Lids	Brows	Pupils	Mouth
LIT · smug	Half-lidded, slanted in	Down at the inner end	Track the nearest player	Smirk
HOT · uh-oh	Open, outer corners droop	Up at the inner end	Track, slightly down	Grimace
CRITICAL · panic	Wide	Up at the inner end	Pinned (×0.7), shaking with the body	Panic (glowing)
Blink	Closed for 0.12 s every 2–4 s while LIT	—	—	—

Moment	Lids	Brows	Pupils	Mouth
Gotcha (a tag)	Narrow, wicked slant	Hard down	On the victim	Grin, 0.6 s
Whee (pass flight)	Wide	Up	Up	Ooh

Rules.

1. **One face for every skin** (§3.6). Skins never paint features on the front face zone.
2. The fuse, seams, lamps and stage colours are untouched (GDD §10.4). The face adds at most 0.09 studs at the front.
3. Every part is rigid and only rotates, so the game drives exactly the rig that was reviewed (`art/bomb/face.py`, contact sheet `art/out/Bomb.Face/contact_sheet.png`).
4. Reduce motion (GDD §21.3) stops the pupil shake. The expressions stay, since they carry the timer read.

4. World

4.1 Temporary map (P0–P2): Meadow

Chase Game's Meadow field transfers unchanged as a playable placeholder for the core loop. **It never ships to players.** The launch map replaces it in P3, before any public build.

4.2 Launch map: "The Fireworks Yard"

Brief: the loading yard of an old fireworks factory at golden hour. Two loading docks face each other across a painted concrete yard. Crates of fireworks are stacked everywhere, bunting hangs from the cranes, and the factory chimneys glow behind it all.

Zone	Design	Gameplay contract (`Arena`)
Safe Strips = the two **loading docks**	A raised concrete dock (0.6 studs) with yellow-black bumper edges, roll-up shutter doors behind, and sandbag walls at the sides. A green floodlight strip marks the safe edge; the floor is stencilled "SAFE".	`Zones/StartZone`, `Zones/FinishZone`; the soft wall for Bombers is an invisible collider following the dock edge
The field = **the yard**	Weathered concrete slabs (a trim sheet with expansion joints, oil stains and chalk marks), plus painted hazard lanes that light up during the Field Fuse warning.	`Zones/Field`, 140 × 90 studs TUNE
Field Fuse cords	Two thick fuses run in steel gutters along both long edges, with a spark racing along them.	Driven by FuseController
Cover & lanes	4–6 low firework crate stacks (waist-high, 3 × 3 studs) and 2 hand-trucks. Movement-friendly: rounded corners, no snags.	`Gameplay/` folder; collision boxes simplified
Backdrop (non-playable)	The factory facade with a hand-painted "FUSE VALLEY FIREWORKS Co." sign in neon tubes, two chimneys with a warm glow, a gantry crane with bunting, a water tower, distant hills. Fireworks pop in the sky on round wins.	Out of bounds

Signature moment — field detonation: the hazard lanes flash in sequence from one end, then a rolling chain of small charges pops up through the concrete joints (VFX §5.4). It's the yard "going off".

Post-launch maps (backlog): "Rooftop Pier" (a night boardwalk with neon) and "The Quarry" (sandstone, mine carts).

4.3 Lobby: "The Depot"

A cosy railway goods depot beside the factory:

- **Spawn plaza:** cobbles, string lights, and the **MEGA BOMB clock tower**. The tower is wrapped in a giant spiral fuse, and the spark climbs toward a huge bomb at the top as the next event nears. When the event starts, the top bomb pops in a firework shower.
- **Shop kiosk:** a striped-awning firework stall with the featured items spinning on brass plinths.
- **Quest board:** a corkboard with three printed quest cards and pins.
- **Leaderboard wall:** three chalkboard-framed boards, plus the weekly statue plinths.
- **Fuse Range:** a practice yard with padded dummies on springs, a practice-bomb dispenser and a defuse pad.
- **VIP lounge:** a gold-trimmed balcony above the stall.

4.4 Props kit (shared by the map and lobby)

Every prop is built in the `art/` pipeline (§10) with the materials library:

Prop	Notes
Firework crate (3 sizes)	Wood with stencilled labels, rope handles, fireworks poking out
Rocket bundle	Paper tubes with twisted fuses (decor only)
Hand-truck	Painted steel, rubber wheels
Sandbags	Canvas, stitched
Dock bumper	Rubber, yellow/black stripes
Shutter door	Corrugated, painted, chipped
Floodlight	Steel, with a SpotLight
String lights	Bulbs with warm emissive cores
Bunting	Triangular flags in the palette, animated sway
Spring dummy	A padded canvas body on a coil spring (Fuse Range)

4.5 Lighting

Context	Preset (§2.5)
Rounds	Golden Fuse
MEGA BOMB rounds	Night Show
Last Stand	Spotlight (a tween from the current preset)

5. VFX

All textures are authored in-house (Blender renders or procedural Python generators in `art/vfx/`), stylised with cel-shaded smoke and hand-designed shapes. None are AI-generated, and none are stock.

5.1 The explosion stack — "Classic Kaboom" (default)

t (s)	Layer	Spec
0.00	**Pre-swell**	The bomb scales up 1.25 over 0.08 s (the anticipation beat)
0.08	**Flash**	A white-yellow core sprite (a 512² radial with a star-shaped falloff). Size 9 studs → 0 over 0.06 s. A PointLight (range 26, brightness 6) for 0.15 s
0.08	**Fireball**	A flipbook of 16 frames (a 4×4 sheet, 1024²) of a toon-shaded fireball: 3 tone bands (white core → `Spark` → `Fuse`) with an `Ink` outline edge. 3 particles at 6–8 studs, lifetime 0.35 s
0.08	**Shockwave**	A floor decal ring (a crisp double ring, `Paper` at 60 %) expanding 0 → 14 studs over 0.25 s and fading. Plus an air-ring billboard expanding to 10 studs
0.10	**Sparks**	24 streak particles (velocity-aligned), `Spark` → `Fuse`, speed 40–70, with drag, lifetime 0.4 s
0.10	**Debris**	6 meshes: 4 iron shell shards and 2 brass rivets. Real parts with physics, CanCollide off with the players, bouncing on the floor, gone after 1.2 s
0.15	**Smoke**	8 toon smoke puffs (a flipbook of 8 frames: dark core → light rim, with an outline), rising and expanding to 7 studs, lifetime 1.1 s
0.15	**Soot decal**	A starburst scorch decal on the floor, fading over 6 s
0.30	**Embers**	12 slow, floating `Ember` particles drifting up, lifetime 1.5 s

- **Total:** ≤ 220 particles at peak.
- **The light variant** (more than 3 booms at once): flash + fireball + 8 sparks + 3 smoke puffs.

5.1b The slap (GDD F-BOMB-10)

The tag's beat, in the same toon language as the explosion stack. It plays on the contact frame, `SLAP_LEAD` after the server's Stick.

Layer	What it is
Smack	`art/vfx/make_smack.py`: a 4×4 flipbook (1024², 16 frames, drawn at 4× and filtered down, so its edges are anti-aliased). A comic impact star: 9 uneven spikes, white core → `Spark` → `Fuse` inside a thick `Ink` outline. It pops in with an overshoot, thin speed strokes fly out between the spikes, and it snaps shut into its core. One particle, 2.8 studs, lifetime 0.32 s, on the Runner's back
Flash	The 6-point star flash, 2.6 → 0.6 studs over 0.09 s
Spray	7 velocity-aligned streaks, `Paper` → `Spark`, sprayed the way the slap travels
The hand bomb	The hero bomb at 0.42× in every ready Bomber's right fist: the Bomber's skin, the face watching the nearest Runner, the fuse leaning out so its spark shows. It pops back in (a 0.22 s swell) when the tag is ready again
Sound	Pro Sound Effects "Hand Slap, Sharp Smacking" (4 takes, thud tail trimmed) over the clamp, plus a quick high whoosh on the swing

Layer	What it is
Motion	`tools/anim/Slap` (Bomber, upper body) and `Stumble` (Runner, upper body), in-house v1. The animator's brief has the final versions

5.1c The Hand-Off (GDD §7.4b)

Layer	What it is
Danger Ring	`art/vfx/make_danger_ring.py`: a 1024² ground ring, white, tinted in game to the stage colour. 20 chunky fuse-link segments with rounded ends, each with a crisp `Ink` rim and its own soft `Ink` shadow (it reads on concrete, at night, and on the orange logo); a hairline inner ring, a soft zone tint inside it, and four inward notches like a sight. Drawn at 4× and filtered down. In game it's an **unlit SurfaceGui** (glows the same in sun, shade and LIGHTS OUT), drawn at the true pass reach (8 → 11 studs across), the links crawling faster and the ring pulsing with the bomb's tick as the fuse burns, popping on a stage change or a new Carrier, and stretching into an oval along a lunge
SLAM PASS	Flash + shockwave ring + 9 streaks sprayed the way it was shoved, the slap at low pitch over a low clamp, a 0.1 s freeze
REJECTED!	A bright `Runner`-blue parry flash + ring + 12 streaks at the rejecter, a high "ting" (the in-house tick at 2.6×) over a high clamp, both rigs frozen 0.08 s, and the hero bomb ricocheting off the rejecter and back onto the giver
Whiff	Toon dust (the smoke flipbook, tinted concrete) at the feet, a low miss-whoosh, the Stumble clip
Every pass (D-PASS-02)	The giver plays the slap swing (their own Slap-Pass swings on their screen at once). The bomb flies 0.26 s on a 3.2-stud arc, its ribbon widening with the chain, and the receiver's back bomb appears only as it lands (no second bomb mid-air). **On landing:** a flash + ring + streak burst on the back, growing with the chain and tinted `Spark` → `Ember` (×3) → `Detonate` (×5); the receiver flashes white for 0.16 s and plays the Stumble; a meaty slap under the clamp; both rigs freeze 90 ms + 12 ms per Carrier (max 150 ms). From ×3 a light-ray burst behind the receiver, from ×4 (or a CLOSE ONE) the crowd gasps, from ×5 the presentation slow-mo. **When it lands on you:** your screen edge flashes in the stage colour, the camera punches in harder, and the shake grows with the chain
Reach tether (F-BOMB-15)	While you carry, an arc from your bomb to whoever a Slap-Pass would hit: a solid ribbon core in the stage colour (not additive, so it reads on bright floors and sky) over an `Ink` edge, with a soft additive glow round it, crackling (width and brightness flicker every frame), bowing upward. Only on your own screen. When it shows, the BUMP! arrow hides (one marker at a time)
Chain Song note	`art/audio/make_chain_note.py` → `chain_note.wav`: an in-house marimba-style mallet (modal synthesis: the tuned bar with its ≈4× and ≈10× partials, a felt strike, a resonator body, a short room), tuned to C5 within 0.2 cents. The game plays it up a major scale, one step per pass. Uploaded (`rbxassetid://12031 2887696662`)
×N tag	The chain count over the Carrier from ×2 (`Spark` → `Ember` from ×3 → `Detonate` from ×5), popping in; the bomb's seams glow hotter with every Carrier it's been through

5.2 Fuse spark (default "Classic")

- A core sprite (a 6-point star, flickering size 0.3–0.45) plus 6–10 tiny streak sparks per second, falling with gravity.
- A PointLight (range 4, warm).
- **At CRITICAL:** the rate triples and a sputter pops every 0.2 s.

5.3 Defuse

- **Steam curl:** a 6-frame flipbook rising from the socket.
- A **"snuff" pop**: a small grey puff.

- A soft green ring on the floor (`Safe`) for 0.4 s.
- **Clutch Defuse adds** a gold flash, gold star sparks, and a 0.5 s light ray burst behind the player.

5.4 Field detonation

- The hazard lanes flash in sequence along the field (0.05 s per lane).
- Then a **rolling chain** of 30 small charges pops up through the concrete joints over 0.6 s, each a mini fireball and smoke.
- Plus a dust sheet across the whole field, then a lingering haze for 2 s.

5.5 Feedback VFX

Effect	Spec
Tag-ready hand glow	A soft `Bomber` glow on the Bomber's hands when the cooldown is ready
Pass arc trail	A warm ribbon trail behind the flying bomb, 0.18 s
Ragdoll soot	The flung body gets a soot overlay and drifts small smoke wisps for 3 s
Bomber stun	Cartoon spinning sparks over the head with a soot face for 3 s
Level-up firework	A personal firework above the player's head; the colours follow their equipped Fuse Spark
Win crown	A gold crown with a lit sparkler, rotating slowly
MEGA BOMB	A giant version of the bomb with an animated crown fuse. Its STOMP telegraph ring is `Mega` violet with hazard dashes

5.6 Explosion cosmetics (launch)

Same timing and 7-stud splash ring as the default (GDD §10.4):

Explosion	Rarity	Concept
Classic Kaboom	Default	§5.1
Confetti Cannon	Common	A paper confetti burst with streamers and a party-horn layer
Ink Splat	Rare	A comic ink splash with a floor splat decal; black and `Paper`
Firework Peony	Rare	A spherical firework burst with trailing stars
Pixel Burst	Epic	Voxel cubes bursting and blinking out; an 8-bit pop layer
Frost Nova	Epic	Ice shards and a frost ring that freezes the floor decal briefly
Black Hole	Legendary	A 0.15 s implosion (it sucks particles in) then a violet pop with lensing rings
Champagne	Exclusive (VIP)	A gold foil cork pop, bubbles and gold confetti

6. UI

2026-10-17: the UI follows the owner's reference video (see §2, "Toy Arcade"). The implemented kit is `src/client/UI/Skin.luau` (native UICorner / UIStroke / UIGradient, no textures) with the panel frame in `DailyController.panel`, the dock tiles in `UI/Dock.luau`, and the sounds `UiHover` / `UiPress` / `UiRelease` / `UiDeny` / `UiOpen` / `UiClose` / `UiTab`. The specs below are the original ones; where they differ, the kit wins.

6.1 The look: "Workshop Arcade"

- **Chunky, tactile and lit.** Panels feel like painted enamel signs with brass fittings.
- **Buttons feel like real arcade buttons:** they physically sink when pressed.
- **Everything has an `Ink` outline and a solid offset shadow**, so it reads on any background, bright or dark.

6.2 Panel system (`UIKit.panel`)

Layer (back → front)	Spec
Drop shadow	A solid `Ink` at 45 %, offset 0, +6 px (straight down), with the same corner radius
Outline	A 3 px `Ink` UIStroke
Base fill	`Ink2` (dark panels) or `Paper` (light cards)
Bevel highlight	A 2 px inner top highlight (`Paper` at 12 % on dark, white at 60 % on light)
Surface texture	A subtle enamel grain (a 256² tiled noise at 4 % opacity): it reads as "material", not "flat"
Header plate (optional)	A brass-coloured strip with rivet dots at the corners, holding the title in Luckiest Guy

Corner radius: 14 px on panels, 12 px on buttons, 999 px on pills.

6.3 Buttons (`UIKit.button`)

State	Visual	Motion	Sound
Idle	A 2-tone face (top 55 % lighter): primary = `Fuse` / `Ember`, secondary = `Smoke`. A 6 px base "skirt" in a darker shade under the face	—	—
Hover (PC)	Face +6 % brightness; a 1.03 scale spring	Spring 0.12 s	`UI.Hover` (very soft)
Press	**The face sinks 4 px into the skirt**; scale 0.96	0.06 s down	`UI.Press` (a mechanical click)
Release	Springs back with a small overshoot	Spring (stiffness 400, damping 22)	`UI.Release`
Disabled	`Smoke` face, `Ash` text, no skirt shine	—	`UI.Deny` on tap, with a horizontal shake

6.4 Icon set (custom, `ui/icons/`)

- **Authoring:** hand-drawn vectors (SVG), exported at 2×: 128² and 256² PNG.
- **Style:**
  - a 4 px `Ink` outline
  - a 2-tone fill (base plus a lighter top-left plane)
  - one specular dot
  - a consistent 10° tilt on objects
- **The launch set (≈ 40):**
  - Star (currency)
  - Bomb, Fuse, Spark, Shield (safe), Runner, Bomber, Crown, Flame (streak)
  - Gift, Calendar, Quest scroll, Crate (Spark and Inferno), Trophy
  - Gear (settings), Lock, Check, Arrow, Close, Plus
  - Robux-safe "R$" price badge (our own frame around Roblox's official Robux glyph, per their brand rules)
  - Category icons (Bomb skin, Explosion, Spark, Trail, Victory, Title, Emote)
  - Dash, Slide, Stop (action buttons)
  - Mega
- **No emoji anywhere.** Not in strings, chips, buttons or callouts.

6.5 Motion system (`UIKit.motion`)

Interaction	Spec
Panel open	Scale 0.85 → 1 with a spring (stiffness 320, damping 24); fade 0 → 1 over 0.12 s; content staggers in at 0.03 s per item
Panel close	Scale 1 → 0.92, fade out over 0.12 s (no spring)
Number change	Counts up over 0.4 s (ease-out); a small 1.08 scale bump at the end; a tick sound every few steps
Reward pop	The value flies from its source to the Stars counter in a bezier arc. The counter does a punch scale, then counts up
Attention	A gentle 2 s breathing pulse (1.0 ↔ 1.04) on claimable items (gift, quest done, calendar)
Text	Never scaled continuously; only discrete tweens, so whole-pixel snapping doesn't look choppy (the Chase Game lesson)
Reduced motion (setting)	Springs become fades; no shake, no pulse

6.6 HUD components

Component	Spec
Crossing chip	A pill "CROSSING 3/8" in Builder Sans Heavy with 8 small fuse dots (lit = done)
HUD Field Fuse	A horizontal braided-fuse bar inside a steel gutter frame. The remaining fuse shortens from the left, with a spark (particles) at the burn point. At 5 s the frame flashes hazard stripes; at 3 s it shakes
Team counter	A pill: Runner icon + count (cyan) · divider · Bomber icon + count (red-orange). When a count changes, the number flips like a split-flap display
MEGA BOMB chip	A pill: the Mega bomb face badge + "MEGA BOMB 4:12"; violet accents

Component	Spec
Quest tracker	Collapsed: 3 small progress rings. Expanded: 3 rows (icon, text, progress bar). On a tick: a bar fill plus a +1 pop
Hot Streak flame	An animated flame icon (a 6-frame flipbook) whose size steps with the streak; the multiplier sits on a brass tag
Gift ring	The gift icon inside a progress ring with "3:12" under it; when ready it bounces and shows "CLAIM"
Carrier vignette	A full-screen ImageLabel: a radial red edge texture (soft inside, a hazard-stripe hint at the very edge), pulsing with the tick
BUMP! arrow	A chunky arrow pointing at the nearest pass target with "BUMP!"; it bobs
Defuse chevron	An edge-of-screen green chevron with "2.1 s" (green) or a red "TOO FAR"
Action buttons (mobile)	Circular arcade buttons, 110 px (Dash) and 78 px (Slide, Stop), each with its custom icon. The cooldown shows as a draining ring and a "ready" ping

6.7 Callouts

Tier	Spec
Small	Luckiest Guy 28 px, `Paper` with an `Ink` stroke. Pops in at 1.3 → 1 (spring) and floats up 20 px over 0.8 s, then fades. Shows "+5 XP · +1 ★" in Builder Sans below
Medium	40 px, with a coloured fill by type (gold = defuse, red-orange = revenge, violet = betrayal). A starburst plate behind it rotates slowly. Holds 1.2 s
Big	60 px, full-width with a banner plate. Light rays behind, a camera kick, a choir or brass hit. Holds 1.6 s. Max one at a time

7. Screens & sequences

7.1 Loading screen

- **Background:** dusk sky.
- **Centre:** the logo, "FUSE VALLEY". The F's crossbar is a lit fuse cord and the V cradles the bomb; the fuse spark burns along the lettering.
- **The fuse burns as loading progresses**, and the spark races to the bomb as it finishes.
- **Below:** a rotating tip.
- **Exit:** the bomb "pops" into the game camera with a wipe.

7.2 Results

The headline:

- **Podium:** a 3D stage with the top 3 avatars on brass podiums (1st gets a crown and a confetti cannon).
- **Bomber win:** the MVP stands alone on a stage, with soot smoke and a bomber chant.

Your round:

- A paper "receipt" card prints line by line with a printer sound. Each line pays out in a coin burst.
- Then the XP bar fills, possibly into a level-up.
- Then 2 "next" hook chips (GDD §8.4).

7.3 Level up

- **The background dims.**
- **The level number** smashes in (scale 3 → 1 with a shockwave), and a firework burst plays behind it.
- **The reward card spins in** and flips to reveal the reward.
- **A level-track item** (every 5 levels) gets its own 3D turntable reveal.

7.4 Crate opening

1. **The crate drops** onto a workshop table (a heavy thud with dust).
2. **The crate's fuse lights**, and the camera pushes in.
3. **Tension:** light leaks through the lid seams in the **rarity colour** (the hint). The crate rattles harder.
4. **POP:** the lid blows off with a rarity-coloured burst (confetti-sparks; Legendary adds a light-ray fan and a brass fanfare).
5. **The item rises** on a beam, rotating, with its name plate and rarity banner. Then the buttons: **EQUIP** · **OPEN ANOTHER**.

Skip: tapping during steps 1–3 jumps to step 4. Never force the long version on repeat openings.

7.5 Daily calendar

- **A wall calendar:** 7 paper day cards pinned on a corkboard.
- **The current day pulses.**
- **Claiming:** an ink stamp slams "CLAIMED" with a thud, and the reward flies to the counter.
- **Day 7** is a big gift card showing this month's Streak Bomb.

7.6 Event card (MEGA BOMB)

- **A violet poster:** the Mega Bomber silhouette, a countdown, the rules in 3 lines, and the rewards.
- **Two buttons:**
  - "SUMMON — R$ 99"
  - "GOT IT"

8. Launch cosmetic catalogue

Bomb skins are in §3.7 and explosions in §5.6. Every item passes the Quality Bar with its own review renders.

8.1 Fuse Sparks (8)

These change the fuse spark sprite and the overhead ring's spark. They never change the ring's stage colours.

Spark	Rarity	Concept
Classic	Default	§5.2
Sparkler	Common	A wedding-sparkler burst, long thin streaks
Ember Drift	Common	Lazy floating embers, a low-key glow
Electric	Rare	Blue-white arcs crackling around the tip
Pixie	Rare	A soft pastel glitter that trails
Plasma	Epic	A magenta plasma ball with orbiting motes
Gold Leaf	Legendary	Gold flakes that peel off and flutter down
Insignia	Exclusive (group perk)	The group colours, a star-shaped spark

8.2 Trails (8)

These show on Dash, Slide and the Last Stand sprint. Ribbon Trails, 0.25 s lifetime, never in Runner cyan or Bomber red-orange.

Trail	Rarity	Concept
Soot Streak	Common	Charcoal smoke ribbon
Paper Streamers	Common	Twisting confetti ribbons
Spark Line	Rare	A gold dotted spark line
Rocket Exhaust	Rare	A firework-rocket flame trail with sparkle
Comet	Epic	A star head with a dust tail
Neon Tube	Epic	Bent neon-sign glass, flickering
Aurora	Legendary	Layered aurora curtains
Victory Lap	Exclusive (level 30)	Chequered flag ribbons

8.3 Victory Booms (6)

These play on your podium spot and when you win the Last Stand.

Victory Boom	Rarity	Concept
Firework Volley	Default	Three peony shells over you
Confetti Rain	Common	Confetti cannons from both sides
Golden Fountain	Rare	Ground fountains of gold sparks

Victory Boom	Rarity	Concept
Grand Finale	Epic	A full skyline firework show (5 s)
Dragon Dance	Legendary	A paper dragon of sparks circling you
Champagne Tower	Exclusive (VIP)	Gold bubbles and foil

8.4 Titles (20)

Source	Titles
Level track	Fresh Fuse (starter) · Spark Plug · Short Fuse · Hot Potato · Pass Master · Fuse Whisperer · Demolitionist · Blast Radius · Living Legend · Big Bang
Achievements (stats)	Clutch King/Queen (100 clutches) · Revenge Served (100 revenge passes) · Chain Reaction (25 ×5 chains) · Last One Standing (10 Last Stand wins) · Unboomable (50 wins)
Events & special	Mega Bomber · Mega Survivor · Dethroner · Weekly Champion · VIP

Title plate: the text is shown on a brass plate under the nametag. The plate's rarity frame follows the title's rarity.

8.5 Emotes (8, lobby and podium)

Wave · Fuse Dance · Juggle (juggling three mini bombs) · Bow · Shrug · Victory Pose · Kaboom (mimes an explosion) · Sit & Sip.

All emotes are animated with Chase Game's `tools/anim` baker pipeline (our own clips).

8.6 Level track (exclusives, every 5 levels)

Level	Reward
5	Ember Drift spark
10	Copper Kettle skin (the earned variant: a bronze plate)
15	Title "Pass Master"
20	Spark Line trail
25	**Tempered** bomb skin (exclusive: blued steel with heat-tint rainbow bands)
30	Victory Lap trail
40	**Meteorite** bomb skin (exclusive)
50	**Obsidian Crown** title plus a crown cosmetic

9. Audio

Sources: the Roblox licensed audio library (SFX and music), **layered and processed in-engine**:

- `SoundGroup`s with EQ, compression and reverb
- pitch randomisation
- multi-layer stacks

There are no AI-generated sounds or voices. Every key sound is a stack of 2–4 layers, so ours sounds like no other game.

9.1 Mix groups

Group	Contents	Volume	Effects
Music	Round theme, lobby theme, stingers	0.55	Ducks −6 dB under Big callouts and booms near you
SFX.World	3D sounds: bombs, booms, footsteps	0.9	A shared reverb send (yard)
SFX.Self	Your own bomb, heartbeat	1.0	No reverb, centred
UI	Buttons, rewards	0.7	Light compression
Crowd	The Last Stand crowd	0.6	Reverb

9.2 Registry (`Assets.Sound.*`)

Key	Moment	Layers (design)
`Bomb.Clamp`	A bomb sticks or is passed	A heavy metal clank + a spring "boing" tail + a leather slap
`Bomb.Ignite`	A fresh bomb lights	A match strike + a fuse catch whoosh
`Bomb.Tick`	Each tick	A wooden clock tick + a tiny metal ping (pitch steps by stage)
`Fuse.Hiss`	Loop while lit	A sparkler hiss, filtered; it gets brighter by stage
`Bomb.Critical`	The CRITICAL whine	A rising whistle + a kettle overtone
`Bomb.Whoosh`	Pass flight	A short air whoosh + a doppler
`Boom.Core`	Explosion	A punchy kick-transient + a cartoon boom body + a debris rattle + a long tail
`Boom.Far`	Distant explosion	A low-passed boom tail (more than 40 studs away)
`Fuse.Snuff`	Defuse	A steam hiss cut-off + a "pft"
`Bomb.Release` / `Bomb.Drop`	Defuse clamp, bomb on floor	A latch click / a heavy iron clunk + a roll
`Clutch.Choir`	Clutch Defuse	A choir "aah" hit + a gold shimmer
`Field.Klaxon`	Field Fuse at 5 s	A factory klaxon, 2 pulses
`Field.Detonate`	Field detonation	A rolling firecracker chain + a big low boom

Key	Moment	Layers (design)
`Heartbeat`	Carrier, synced	A low double-thump (SFX.Self)
`Crowd.Chant` / `Crowd.Cheer` / `Crowd.Groan`	Last Stand	Stadium crowd beds
`Stinger.Bombers` / `Stinger.Victory` / `Stinger.BombersWin`	Phase beats	Brass and percussion hits
`UI.Hover` / `Press` / `Release` / `Deny`	Buttons	Mechanical arcade switch clicks; deny = a muted buzz
`Reward.Coin` / `Reward.Level` / `Reward.Crate*`	Rewards	Coin chimes (pitch-up per coin), a level fanfare, crate rattle/pop by rarity

9.3 Music

Track	Use
Lobby theme	A relaxed, swung, upbeat tune (licensed library)
Round theme	Driving percussion. The escalation (GDD §10.1) is done with SoundGroup filters, a hi-hat layer, a drum cut and the heartbeat layer
MEGA BOMB theme	A heavier version for event rounds
Last Stand	The drumroll bed, plus the crowd

10. Pipeline

10.1 Folder layout (new repo)

```
art/
  lib/            materials.py, shapes.py, kit.py (bevels, trims), render.py (turntables), export.py
  bomb/           build.py (the hero bomb), skins/<Skin>.py
  props/          one build.py per prop family
  vfx/            texture generators (flipbooks, sprites) + Blender sims rendered to sheets
  out/<Asset>/    FBX, .blend, textures (PNG), manifest.json, contact sheet
ui/
  icons/          SVG sources + exported PNG @2x
  textures/       panel grain, vignette, ring masks
audio/            layer recipes (JSON): which library sounds, pitch, gain, effects
```

10.2 Authoring rules

1. **Deterministic scripts.**
   - Every 3D asset is a Blender 4.5 Python build script, so it can be rebuilt and tweaked by parameters.
   - This is the same approach as Chase Game's `art3d` library, which proved that it works.
   - Run with `blender -b --python-exit-code 1 --python art/<path>/build.py`.
2. **Real modelling:**
   - Bevels on every hard edge, supporting loops and smooth shading.
   - Detail goes in the geometry where it affects the silhouette, and in the normal map where it doesn't.
3. **Textures are baked in Blender** from procedural materials into 1024² (hero) or 512² (props) PNG sets: colour, normal, roughness, metalness.
4. **Scale:** 1 Blender unit = 1 stud. Check against the 5-stud R6 reference figure in every render.
5. **Orientation:** export as Roblox −Z forward, +Y up. Pivots go where the gameplay needs them (the bomb's pivot is the clamp mount).

10.3 Review (the gate for "premium")

Every asset produces a **contact sheet**:

- the hero shot
- 8 turntable angles
- a material close-up
- a black silhouette
- the R6 scale view
- **for the bomb:** all 3 ticking stages, plus on-back views at 6, 25 and 60 studs

Rendered with Cycles under the three lighting presets.

A hero asset doesn't move to integration until the user approves its sheet.

10.4 Import to Roblox

Asset type	Process
Meshes	Studio 3D Importer: Merge Meshes **off**, pivot to scene origin, Scale Unit = Studs; verify against `manifest.json`

Asset type	Process
Textures	SurfaceAppearance (ColorMap, NormalMap, RoughnessMap, MetalnessMap), uploaded to the group
UI images	Uploaded to the group; the ids go in `Assets.luau`
Animations	Chase Game's `tools/anim` baker; published to the group by the user

Everything is referenced only through `Shared/Assets.luau` keys.

10.5 Naming

`<Category>.<Name>[.<Variant>]`, for example:

- `Bomb.ClassicIron`
- `Bomb.Skin.Delft`
- `Prop.Crate.Large`
- `Vfx.Fireball.Classic`
- `Icon.Star`
- `Sound.Boom.Core`

Code, manifests, review sheets and the store catalogue all use the same key.